Assignment 2

Cameras and Lenses

Instruction: This assignment has 2 part of lab problems including part 1 is asking about cameras and lenses and for part 2 is setting camera parameters. Please try to answer all questions in English.

Part 1 : Cameras and Lenses

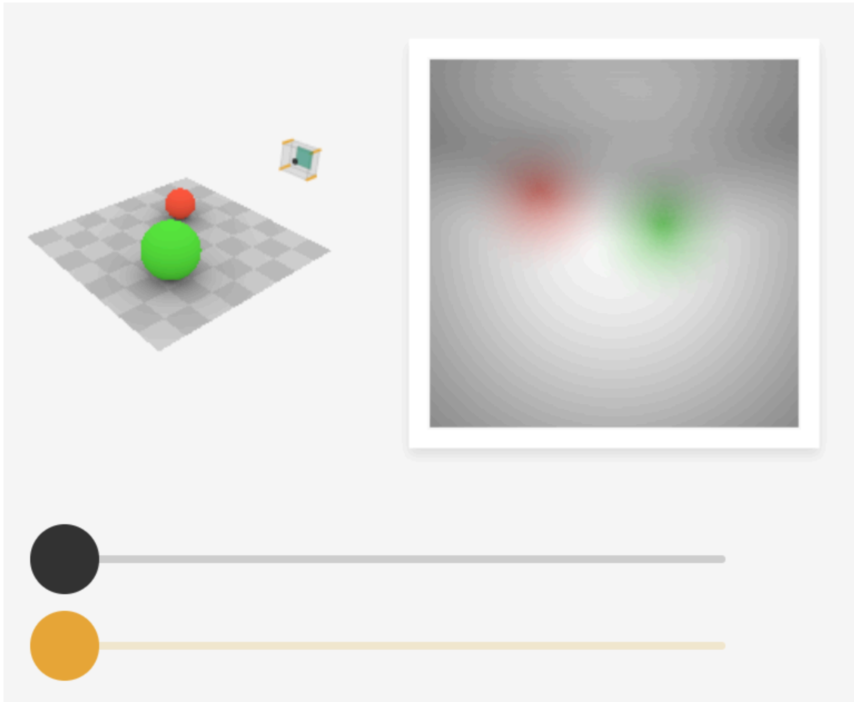
1.1 Exploring Pinhole Camera Configurations: The Effects of Hole Diameter and Sensor Distance on Image Formation

Question 1: How does the size/diameter of the pinhole affect image sharpness and brightness?

Ans: size of pinhole is relate to brightness and sharpness because when size is bigger that can have more space for receive light form object.

Question 2: How does the distance between the opening and the sensor change the size of objects and the field of view?

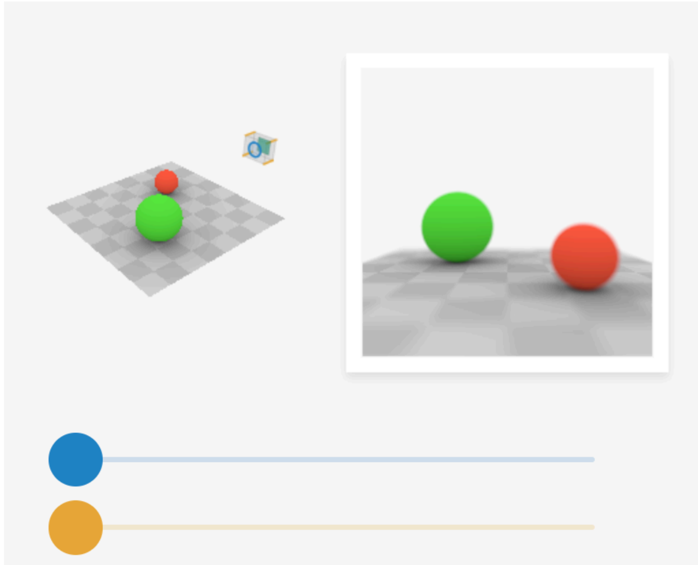
Ans: the distance between the opening and the sensor ,Increasing the distance, field of view becomes narrower and object in the image appear larger.



1.2 Camera Configurations

Question 3: What happens to the image when we adjust the distance between the lens and the sensor? How does it affect focus and image clarity?

Ans: If the distance between the lens and the sensor does not match the focal length,the image will look blurry. The limit of lens is you need to match the focal length with appropriate distance between the sensor and lens to make the image clearly.



Question 4: When the object (source) moves further away from the lens:

- Does the image's position become closer to or further from the lens?
- Does the size of the object's image on the sensor become larger or smaller?

$$\frac{1}{S_o} + \frac{1}{S_i} = \frac{1}{f}$$

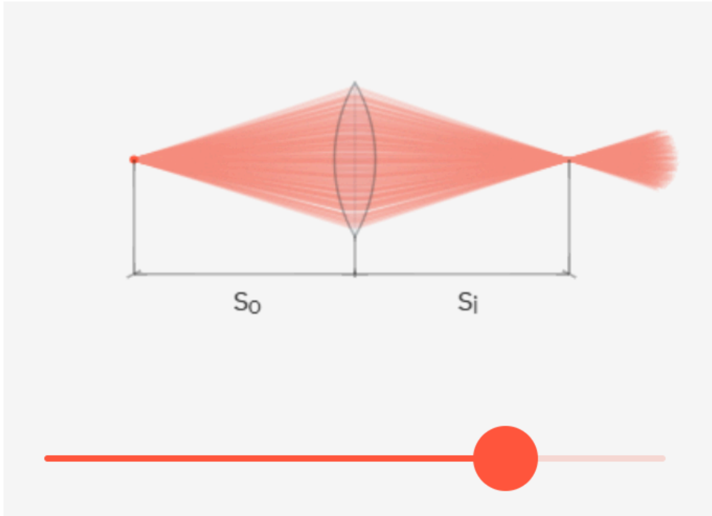
$$M = \frac{S_i}{S_o}$$

S_o: Distance between the object and the lens

S_i: Distance between the image and the lens

f : Focal length

M : Magnification



Question 4: When the object (source) moves further away from the lens:

Does the image's position become closer to or further from the lens?

Ans the image’s position becomes closer to the lens.

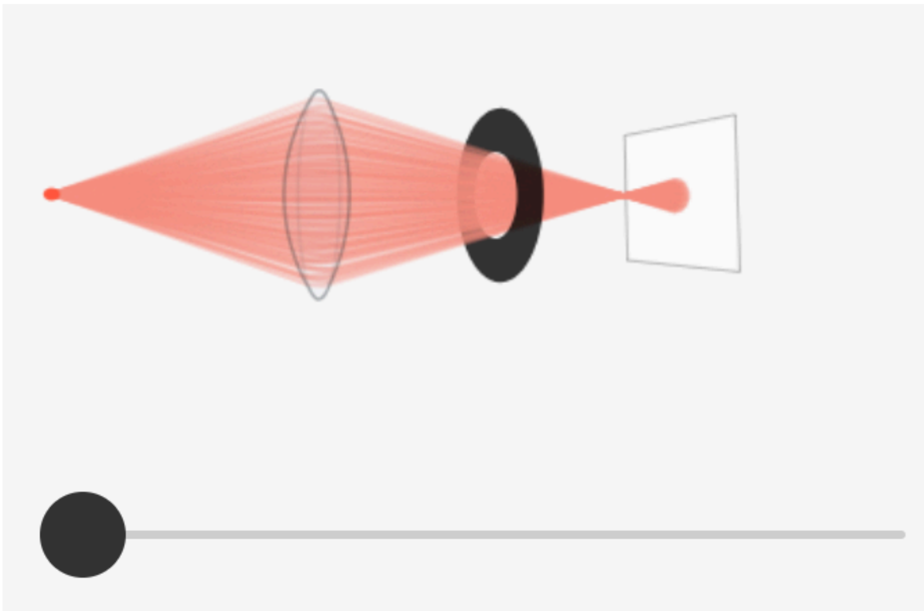
Does the size of the object’s image on the sensor become larger or smaller?

Ans size of the object’s image on the sensor becomes smaller.

Question 5: What is depth of field?

Ans The distance between the nearest and the farthest objects that are in acceptably sharp focus in an image which can be determined by focal length.

Question 6**:** How does decreasing the aperture size affect the depth of field?



Ans The smaller the diameter of aperture, the further the focal length(which determines the depth of field)

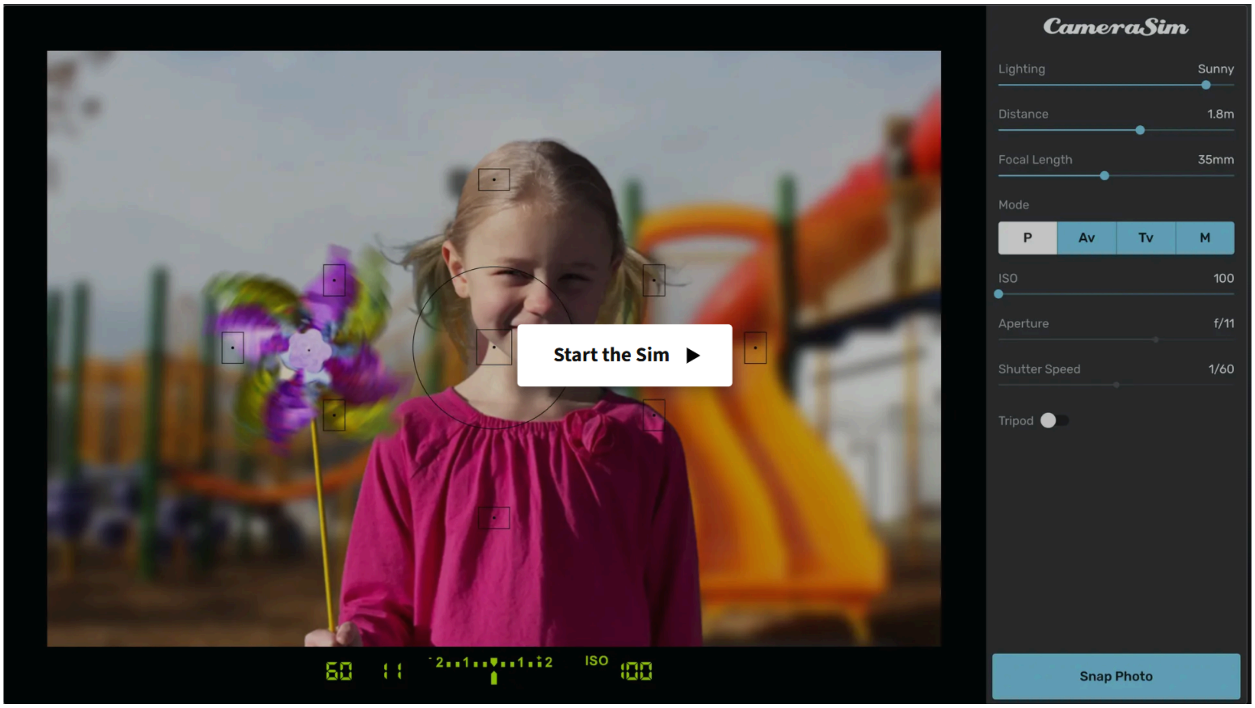
To explore and experiment with the effects, you can refer to the interactive simulations available in the link below. Use the provided controls to find answers and enhance your understanding.

https://ciechanow.ski/cameras-and-lenses/

Part 2 : Exposure Triangle

Instruction : For 2nd challenges you'll try to adjust 3 of factors (aperture, shutter speed, ISO) for taking photo of little girl through virtual camera lenses.

Virtual camera lenses : https://www.camerasim.com/original-camerasim



Default environment setting :

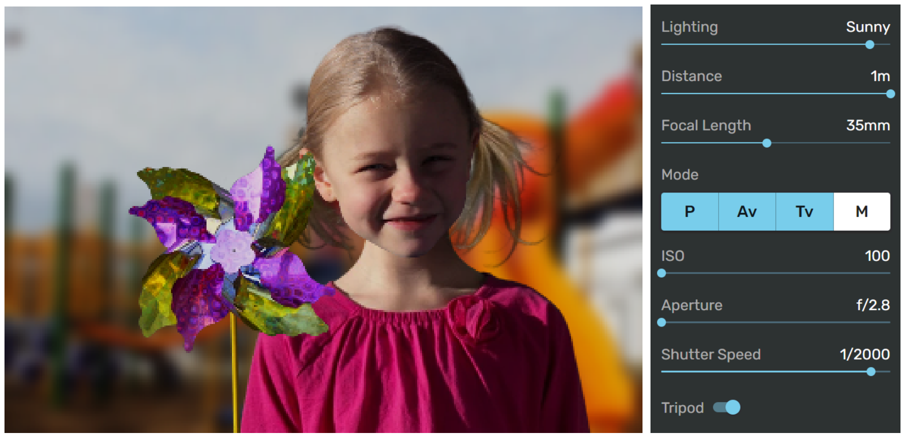
Distance : 2m.

Focal Length : 35mm.

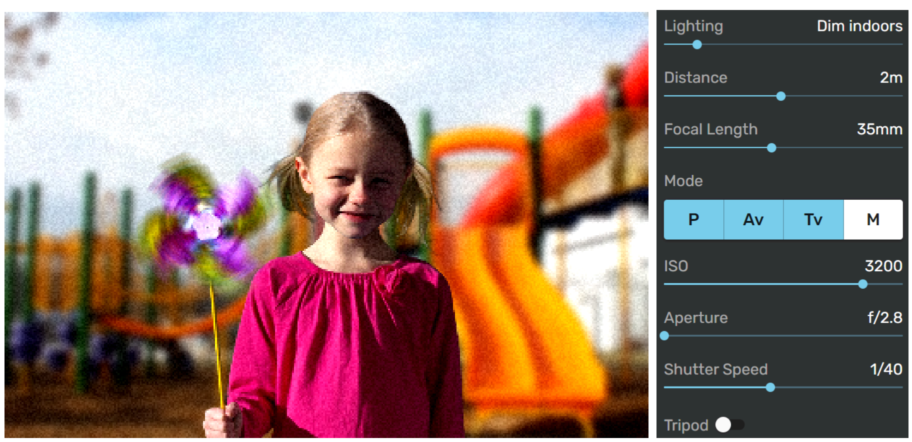
Challenges problem :

1. Try to adjust camera parameters and get snap photo result that showing the best looking of the girl with blur background and the lowest noise.

1.1 Set Linghting: Sunny

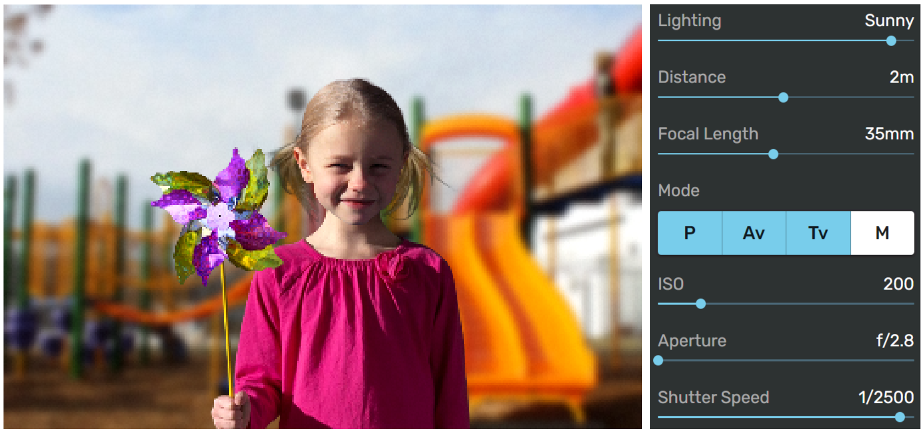


1.2 Set Linghting: Dim indoors



2. Try to adjust camera parameters and get snap photo result that showing stop the fan motion with the lowest noise photo.

2.1 Set Linghting: Sunny



2.2 Set Linghting: Partly cloudy

2.3 Set Linghting: Bright indoors